

NextGen Solutions Group Project

Computer Network Lab — Group Project

Design and Implementation of “NextGen Solutions Pvt. Ltd.” Network

1. Introduction

You have learned how to build a basic network, configure routers, assign IP addresses using FLSM and VLSM, set up DHCP, and connect networks using Static Routing and RIP. It is now time to bring everything together and build one **complete, working corporate network**.

This is a **group project**. Every group will build the **same company network** described below, with the **same departments** and the **same topology**. The only thing your group decides for itself is **how many hosts (PCs) each department has**. Because of this, your final IP addressing and configuration will still end up a little different from other groups, even though the scenario is identical for everyone.

2. The Scenario (Fixed — Same for All Groups)

Company: NextGen Solutions Pvt. Ltd.

The company has 5 departments, and each department is connected to the network through its **own dedicated router**. One department — the IT/Server Department — sits in the middle of the chain and acts as the core, hosting the shared servers. **No router in this network may use more than 3 interfaces**.

Department	Role in the Network
HR Department	Regular LAN, connects to the core
Finance Department	Regular LAN, connects to the core
IT / Server Department	Core department. Hosts the DHCP Server and the HTTP Server.
Software Development Department	Regular LAN, connects to the core
Sales & Marketing Department	Regular LAN, connects to the core

3. Fixed Topology (Same for All Groups)

You must use **exactly 5 routers**, arranged in a **linear chain**, with the IT/Server Department in the middle. This arrangement is required so that **no router needs more than 3 interfaces** (each router uses at most 1 LAN interface + 2 serial interfaces to its neighbors):

HR Dept (Router R1)	----	Finance Dept (Router R2)	----	IT/Server Dept (Router R3, CORE)	----	Software Dev Dept (Router R4)	----	Sales & Marketing Dept (Router R5)
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- Each router connects to its **own department LAN** through a switch (1 interface), exactly like your earlier labs.
- Each router also connects to its **neighbor router(s) in the chain** using point-to-point serial links (1 interface per neighbor).
- R1** and **R5** are at the two ends of the chain, so they only need **2 interfaces** each (1 LAN + 1 serial link).
- R2, R3, and R4** are in the middle, so they need **3 interfaces** each (1 LAN + 2 serial links) — this is the maximum allowed for any router in this project.
- R3 (IT/Server Department)** sits in the middle of the chain and has its own local LAN, where the **DHCP Server** and **HTTP Server** are physically placed.

Fixed routing assignment (do not change this):

Link	Routing Method
R1 (HR) ↔ R2 (Finance)	RIP/OSPF
R2 (Finance) ↔ R3 (IT/Server, Core)	Static Routing

R3 (IT/Server, Core) ↔ R4 (Software Dev)	RIP/OSPF
R4 (Software Dev) ↔ R5 (Sales & Marketing)	RIP/OSPF

Note on static routing: Since R1 and R3 are not directly connected, reaching R1's or R3's subnets from the "static side" of the network requires more than one static route per router (route the traffic hop by hop through R2). This is intentional — it gives you real practice in chaining static routes across multiple routers, not just one link.

Fixed DHCP assignment (do not change this):

Department	DHCP Method
HR	Router-based DHCP (ip dhcp pool on R1)
Finance	Router-based DHCP (ip dhcp pool on R2)
Software Development	DHCP Server (dedicated server in IT dept, reached via ip helper-address since it is in a different subnet)
Sales & Marketing	DHCP Server (same server, same relay method)
IT/Server Department	Static IP addressing (servers should not use DHCP)

4. What Your Group Decides (The Only Open Part)

Your group must choose **how many hosts (PCs) go in each of the 5 departments**. Choose any reasonable number between **5 and 40** for each department — and it does not need to be the same number in every department.

Example (this is just an example, do not copy it): - HR: 12 hosts - Finance: 8 hosts - IT/Server: 6 hosts (plus the two servers) - Software Development: 25 hosts - Sales & Marketing: 15 hosts

Your host counts will decide how you divide your assigned IP block using **VLSM** — this is why no two groups will end up with an identical addressing table, even though everyone is building the same company network.

Your instructor will assign your group a unique base network address (for example, 192.168.<group_number>.0/24). All subnetting must be done starting from this base block.

5. Mandatory Technical Requirements

Every group must implement all of the following, exactly as specified in Sections 2 and 3:

5.1 Topology

- Exactly 5 routers (R1-R5) in the linear chain shown above, with no router using more than 3 interfaces.
- Each department on its own switch and its own subnet.

5.2 IP Addressing (VLSM)

- Subnet your assigned base block using **VLSM**, based on the host counts your group chose.
- Prepare a complete **IP addressing table**: subnet, network address, usable host range, broadcast address, subnet mask, and which department/link it belongs to.

5.3 Router Configuration

- Hostnames matching the department (e.g., R1-HR, R3-CORE, etc.), console/enable passwords, and a login banner on every router.
- All interfaces correctly addressed and activated.

5.4 DHCP (Both Methods, as Assigned in Section 3)

- Router-based DHCP working correctly for HR and Finance.
- Server-based DHCP (with ip helper-address relay) working correctly for Software Development and Sales & Marketing.

5.5 Routing (As Assigned in Section 3)

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5.6.HTTP Server

The HTTP Server is placed in the IT/Server Department (as shown in the topology).

Host a simple company webpage on it (you may edit the default HTML and add your group name). Demonstrate

- that PCs from **at least three different departments** can successfully load this webpage in a browser inside
- Packet Tracer.

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5.7 OSPF — Optional (Bonus Only)

OSPF is **not required**. If your group wants to try replacing RIP v2 with OSPF on the Software Development and/or Sales & Marketing links purely to explore and compare, you may do so **in addition to** your working RIP v2

- configuration, for bonus marks only. Do not remove the required RIP v2 setup to do this.

6. Deliverables

1. **Packet Tracer file (.pkt)** — fully working, saved with your group name/number.
3. **Group Presentation** — every group must present and demonstrate their project live in the lab (see Section 7).
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7. Final Presentation Requirement

Every group must give a **live presentation and demonstration** of their project to the instructor (and, if scheduled, the rest of the class). The presentation must include:

- A short walkthrough of your topology and your chosen host counts/subnetting.
- A live demo showing: DHCP working on a client, a successful ping across two different departments, and the HTTP page loading in a browser.
- Every group member must be prepared to answer questions about **any part** of the configuration — not only the part they personally typed in. Marks are given per student based on their individual answers during the presentation, not only per group.

8. Evaluation Rubric (Total: 100 marks + 5 bonus)

Criteria	Description	Marks
1. Topology Compliance	Exactly 5 routers, correct linear chain topology, no router using more than 3 interfaces, correct department mapping	10
2. IP Addressing (VLSM)	Correct subnetting for chosen host counts, efficient address use, complete table	15
3. Router & Switch Configuration	Hostnames, passwords, banners, interface configuration, correct activation	10
4. DHCP Server	Router-based DHCP (HR, Finance) + Server-based DHCP with relay (Software Dev, Sales) all working	15
DHCP Router	Correct static routes on HR and Finance links, verified reachability	10
6. RIP/OSPF	Correct RIP v2 on Software Dev and Sales links, routing table verified	10
7. HTTP Server	Server configured correctly, webpage	10

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3. Router & Switch Configuration	Hostnames, passwords, banners, interface configuration, correct activation	10
4. DHCP Server	Router-based DHCP (HR, Finance) + Server-based DHCP with relay (Software Dev, Sales) all working	15
5. Static Routing	Correct static routes on HR and Finance links, verified reachability	10
6. RIP v2 Configuration	Correct RIP v2 on Software Dev and Sales links, routing table verified	10
7. HTTP Server	Server configured correctly, webpage	10